Student Homework Correction & Report

# ADAM BARDANA

| | | |
|---|---|---|
| **Subject:** IGCSE Mathematics | **Topic:** Order of Operations (BIDMAS) | **Score:** 5 / 10 (50%) |
| **Instructor:** Mr. Haitham Hussien | **Chapter:** Chapter 1 (Numbers) | **Grade: D (Needs Improvement)** |

## 1. Question-by-Question Marking Breakdown

| Q# | Topic / Expression | Mark | Status | Student Answer / Error Analysis |
|---|---|---|---|---|
| Q1 | 8 + 12 ÷ 4 = 5 (Explain error) | 1/1 | CORRECT | Correct explanation given. |
| Q2 | 5 × (4² + 2 - 12) | 1/1 | CORRECT | Correct evaluation = 30. |
| Q3 | 5² + √121 | 1/1 | CORRECT | Correct evaluation = 36. |
| Q4 | (23 - 3 × 3)² | 2/2 | CORRECT | Correct evaluation = 196. |
| Q5a | 4 + 9 × 6 - 4 = 22 | 0/1 | INCORRECT | Brackets missing/unattempted. |
| Q5b | 24 ÷ 12 - 8 + 2 = 4 | 0/1 | INCORRECT | Brackets missing/unattempted. |
| Q6a | 10 - 4 ÷ 2 + 18 = 21 | 0/1 | INCORRECT | Brackets missing/unattempted. |
| Q6b | 7 × 3 + 1 + 2 = 30 | 0/1 | INCORRECT | Brackets missing/unattempted. |
| Q7 | 2 + 8(40 - 5) | 0/1 | INCORRECT | Mistake: Evaluated 2 + 8 = 10 first, then 10 × 35 = 350. Addition performed before multiplication (BIDMAS violation). |

## 2. Diagnostic Review & Mistake Analysis

Adam scored 5/10. He demonstrated good competence on straightforward evaluations (Q1-Q4), but made a fundamental BIDMAS error in Q7 by adding 2 + 8 before multiplying by 35. Additionally, he left Q5 and Q6 without brackets.

## 3. Personalized Action Plan & Next Steps

- **BIDMAS Hierarchy Drill:** Re-learn rule: Multiplication ALWAYS precedes Addition (2 + 8 × 35 = 2 + 280 = 282).
- **Bracket Insertion Method:** Practice trial-and-error bracket placement strategies for equations.